



DISCOVER

Start Here

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SEMICONDUCTOR

Semiconductor Recipes

DEVELOP

AI Development Guide



OWNED & SUPPORTED BY

NiceGUI Base



Support



Documentation

VERSION

NiceGUI Base 3.0.0a8

NiceGUI 3.15.0

REFERENCE

Start Here

Start from the engineering intent, visually compare the governed authorities, then open only the reference you need.

Search

⌘K / Ctrl+K



Theme · System

Appearance

NICEGUI BASE · DEPARTMENT STANDARD

What are you building?

Start from intent. NiceGUI Base will lead you to the governed pattern, layout, visualization, component, recipe and scaffold instead of making you browse hundreds of references one by one.

10

Patterns

58

Analytics

34

Components

8

Recipes

Choose an intent

See the recommended pattern, recipe, scaffold, and related governed references before leaving this page.



MONITOR

Monitor a process

Health, trends, limits, alerts and review queues.

Recurring health and exception review



INVESTIGATE

Investigate an excursion

Affected/control evidence, commonality and drill-down.

Evidence-led root-cause investigation



COMPARE

Compare tools or populations

Aligned distributions, deltas and supporting records.

Side-by-side population or tool decisions



EXPLORE

Explore engineering data

Filter, inspect, chart and drill into arbitrary records.

Filter-first record exploration



DASHBOARD

Build an operational dashboard

KPI hierarchy, trends and exception-focused detail.

KPI, trend and exception overview



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REFERENCE

Developer Scaffolding

This saved-link route now points developers to deterministic pattern and recipe starters.

Search

⌘K / Ctrl+K



Theme · System

Appearance

Compatibility authoring surface

Manual composition is retired from the normal reference journey. Start from a canonical pattern or recipe with the installed CLI; domain logic stays in your application services.

CLI · CREATE-PATTERN

Choose a pattern

Use a registered page hierarchy and responsive slot contract.

CLI · CREATE-RECIPE

Choose a recipe

Use a bounded semiconductor workflow and its data contract.

AGENT WORKFLOW

Read the agent path

Resolve a requirement into an authority before coding.



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REFERENCE

Design System

The deterministic visual contract used by the reference explorer, framework components, and generated applications.

Search

⌘K / Ctrl+K



Theme · System

Appearance

Design System Authority

See the token in use first; the compact contract beside each specimen is the source for Python consumers, components, CSS, and generated applications.

Import from nicegui_base.design; choose semantic roles before extending the framework.

Spacing scale

Use for recurring distances; prefer semantic gaps for composition.

LIVE USE



Tight

--cui-space-1 · 4px



Related

--cui-space-2 · 8px



Stack

--cui-space-4 · 16px



Content

--cui-space-6 · 24px



Section

--cui-space-10 · 40px



Page

--cui-space-16 · 64px

classes("cui-stack cui-gap-content")

Copy semantic API

RECOMMENDED USE

Semantic gaps

Use for page, section, content, stack, cluster, and control relationships.

LIVE USE

Page

Section

Content

Control

page_section(..., gap="section")

Copy semantic API

RECOMMENDED USE

Use for page, section, content, stack, cluster, and control relationships.

Don't: invent a local value when a token exists.

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REFERENCE

Settings

Manage appearance and Explorer identity without leaving the department reference shell.

Search

⌘K / Ctrl+K



Theme · System

Appearance

Appearance

These preferences apply to the Explorer and its mounted reference previews. System follows the operating-system color scheme.

Theme

System

Light

Dark

Density

Comfort

Compact

Dense

Motion

Normal

Reduced

Account

Identity is read from the company environment and is display-only here.

AccessKey

Company identity value

Member

Role

Reference Explorer display value

Member

Department

Reference Explorer display value

Department unavailable

About this reference

Use the canonical registries, patterns, tables, visualizations and recipes shown here in application code.

NiceGUI Base 3.0.0a8

NiceGUI 3.15.0

Reference Explorer

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NiceGUI Base 3.0.0a8

NiceGUI 3.15.0

REFERENCE

Design System

One discoverable inventory over canonical NiceGUI Base registries, grouped by the job each capability solves.

Search⌘K / Ctrl+K

Theme · System

Appearance

Search catalog

Name, tag, intent, data shape, dc

Refine results · type, family, intent, data, domain, relation

Analytics Studio

465 canonical Reference Explorer entries are searchable. Choose a family or search to inspect individual capabilities.

Shell/Layout

1 canonical entry

Examples: Layout

Browse family

Controls

35 canonical entries

Examples: ButtonGroup, SplitButton, Divider

Browse family

Forms/Overlays

25 canonical entries

Examples: Form Filter Overlay, Form, FilterBar

Browse family

Tables

7 canonical entries

Examples: Table, DataTable, ServerDataTable

Browse family

Content/Workflow

25 canonical entries

Examples: Content, Jobs, MetricCard

Browse family

Generic Visualizations

25 canonical entries

Examples: Visualization, AreaChart, BarChart

Browse family

Engineering Components

Generic Application Patterns



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REFERENCE

Components

Scan governed component specimens first; open a detail only for deeper states, configuration, accessibility, or code.

Search ⌘K / Ctrl+K

Theme · System

Appearance

ButtonGroup

Interactive specimen · Use the real control below; interactions are recorded in Event / debug output.

Run Pause Reset

ACTIONS · LIVE

ButtonGroup

Group related actions without visual fragmentation

Best for · toolbar actions · mode actions

Open example

Favorite

Compare

SplitButton

Interactive specimen · Use the real control below; interactions are recorded in Event / debug output.

Run Pause Reset

ACTIONS · LIVE

SplitButton

Primary action with closely related alternatives

Best for · export alternatives · run options

Open example

Favorite

Compare

Divider

Interactive specimen · Use the real control below; interactions are recorded in Event / debug output.

Measurement summary

Run details

SURFACES · LIVE

Divider

Subtle structural separation

Best for · sections · menus

Open example

Favorite

Compare

CollapsiblePanel

Interactive specimen · Use the real control below; interactions are recorded in Event / debug output.

Advanced settings

Secondary configuration state is correct.

Accordion

Interactive specimen · Use the real control below; interactions are recorded in Event / debug output.

Scrolling

Every 100 pixels

No limitation

Chip

Interactive specimen · Use the real control below; interactions are recorded in Event / debug output.

Run Pause Reset

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Application Patterns

Choose a finished application structure, see the real governed composition, then adapt business logic in your own services.

Search⌘K / Ctrl+K

Theme · System

Appearance

LIVE GOVERNED COMPOSITION

Dashboard

kpis · trend · exceptions

Dashboard

At-a-glance process performance with trends and exceptions.

Lots today

128

↑ +6%

Across three qualified tools

In control

96.9%

↑ +0.8 pp

Open exceptions

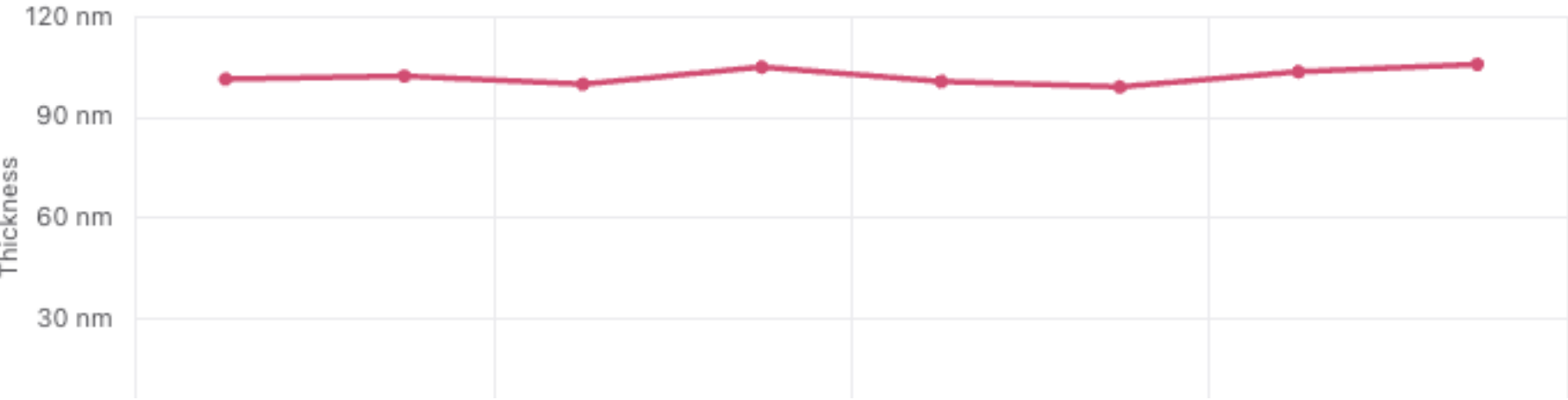
3

Two require engineer review

Thickness trend — last eight lots

+

−



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REFERENCE

Visualizations

See what all 58 governed analytical surfaces actually look like before opening a full interactive reference.

Search

⌘K / Ctrl+K

Theme · System

Appearance



SPC · LIVE

SPC I-MR

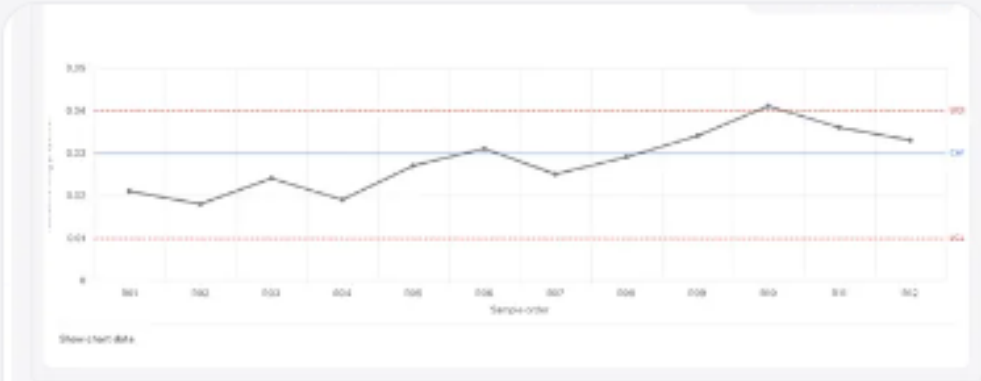
Individuals and moving-range control chart

Best for · continuous measurements without rational subgroups

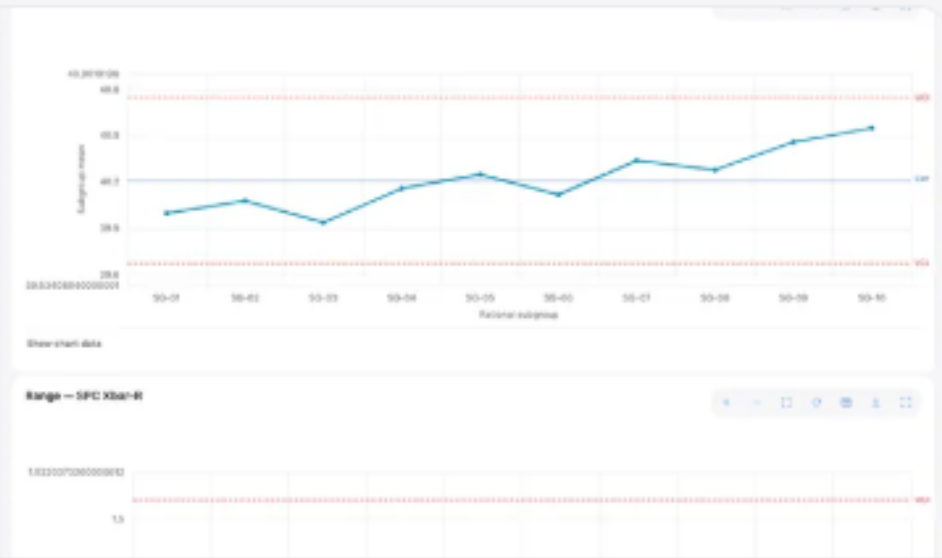
→ Open example

☆ Favorite

Compare



ANALYTIC · SPC
SPC P



SPC · LIVE

SPC Xbar-R

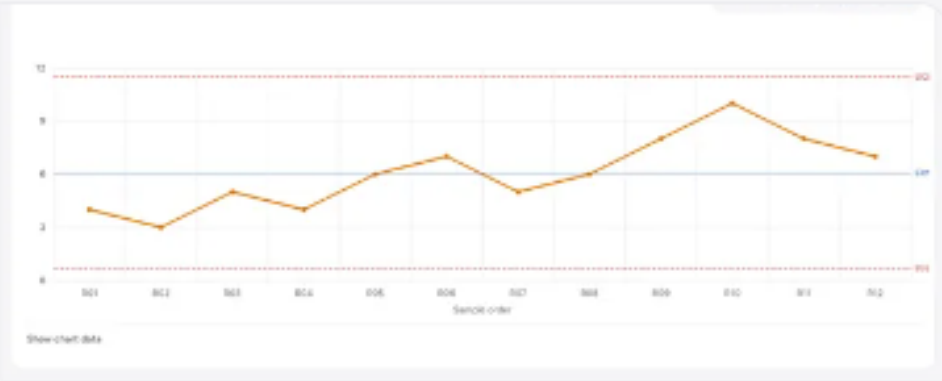
Xbar-R subgroup control chart

Best for · small equal-size rational subgroups

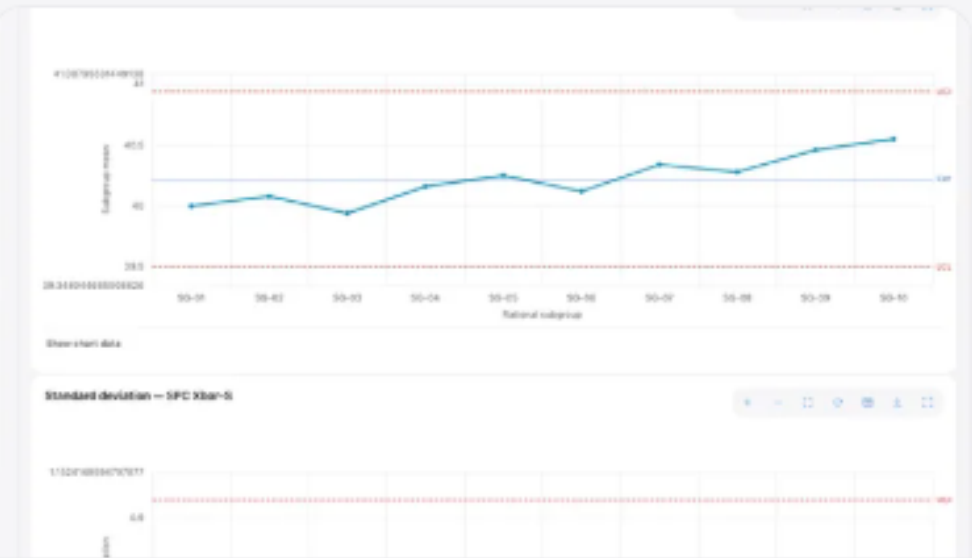
→ Open example

☆ Favorite

Compare



ANALYTIC · SPC
SPC Np



SPC · LIVE

SPC Xbar-S

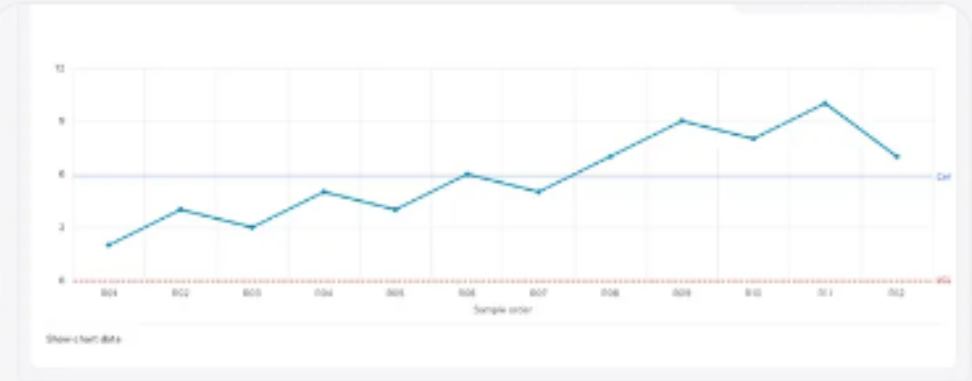
Xbar-S subgroup control chart

Best for · equal-size subgroup variability matters

→ Open example

☆ Favorite

Compare



ANALYTIC · SPC
SPC C

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REFERENCE

Semiconductor Recipes

Scan the domain workflows visually, compare their contracts, then open the recipe that matches the engineering question.

Search ⌘K / Ctrl+K

Theme · System

Appearance



SEMICONDUCTOR RECIPE · LIVE

SPCMonitor

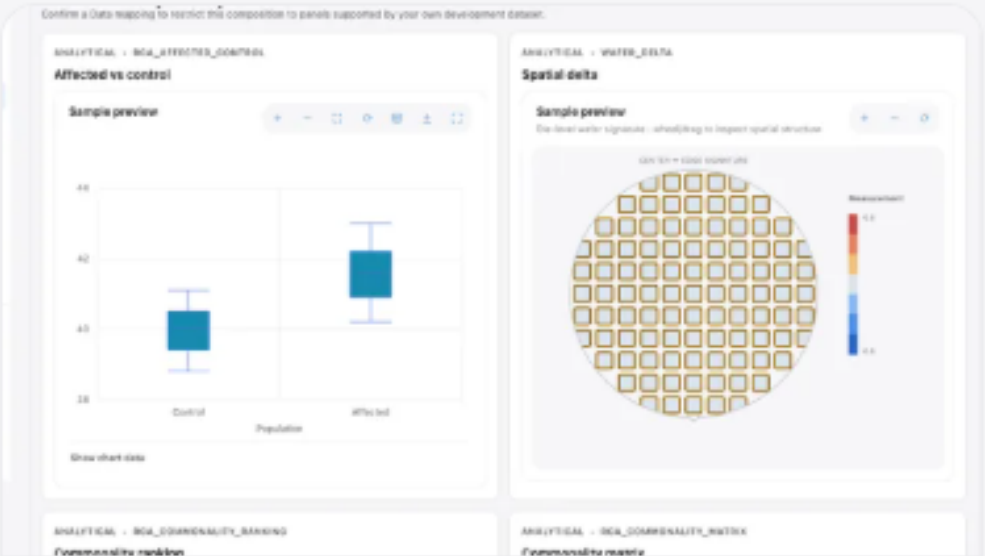
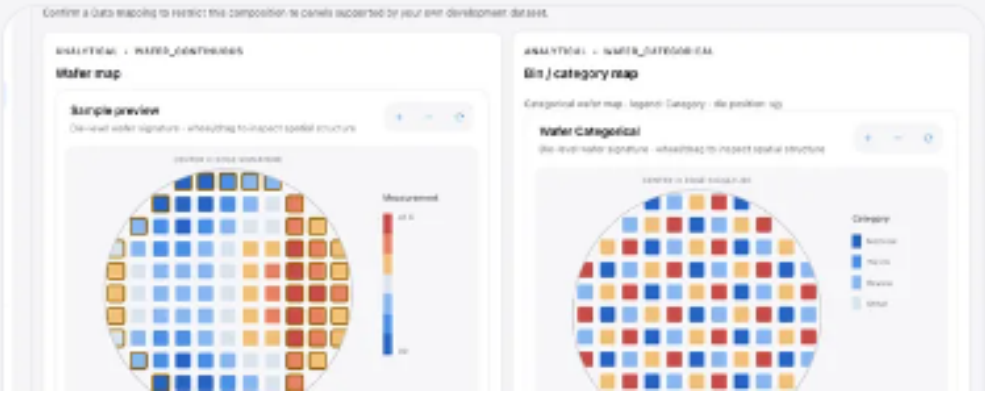
Monitor process stability, capability and violations with drill-down to manufacturing context.

Best for · monitor a process parameter · detect process drift or control-rule violations

→ Open example

☆ Favorite

Compare



SEMICONDUCTOR RECIPE · LIVE

ExcursionDefenseLine

Triage an excursion with affected/control populations, spatial localization, commonality and genealogy evidence.

Best for · build an excursion defense line · triage affected versus control material

→ Open example

☆ Favorite

Compare



SEMICONDUCTOR RECIPE · LIVE

FdcToolHealth

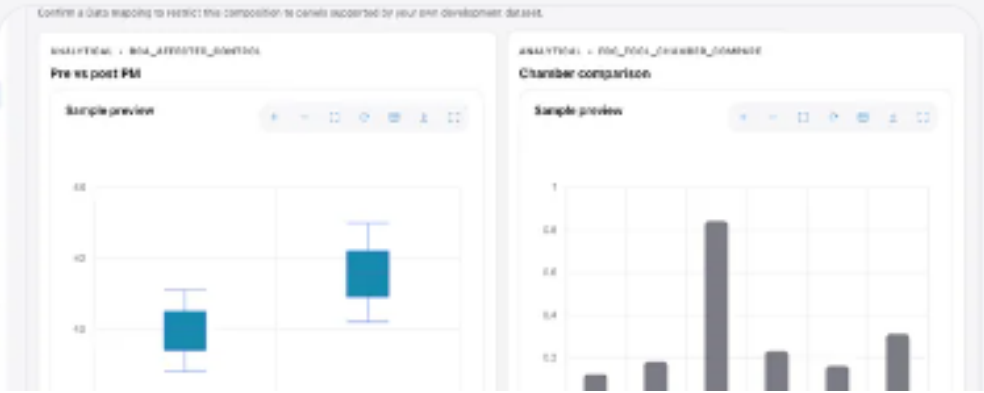
Diagnose equipment health from aligned sensor traces, envelopes, events and multivariate fingerprints.

Best for · analyze FDC traces · check tool or chamber health

→ Open example

☆ Favorite

Compare





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REFERENCE



Data & Tables

A governed production reference for tables, record management, data exploration and provider-scale grids.

Search

⌘K / Ctrl+K



Theme · System

Appearance

Production table lab

The governed production reference for tables, record management, data exploration and provider-scale grids.
One active lab keeps the page fast; every lab reuses the same semiconductor data contract.

Grid

Actions

Edit

Visualize

Import

Detail

Server

States

Grid

The production read-oriented grid: query, layout, views and persisted state.

64 records

Persisted view

Default

Watch / OOS

Unreviewed

Clear filters

Clear sorting

Reset view

64

Records

49.820 nm

Mean measurement

94.1%

Mean yield

8

OOS

View: default · 64 records

Lot and wafer monitoring

Search, filter, sort and save the view you want to reuse.



Search records



Columns



Compact



Export



Refresh



Record



Timestamp



Lot



Waf...



Tool



Cha...



Operation



Recipe



Measurem...



Targ

Actions

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Diagnostics

Understand runtime health, framework coverage, and the few actions that need attention.

Search ⌘K / Ctrl+K

Theme · System

Appearance

System health

A quick answer to whether the installed reference is ready to use.

Reference Explorer

Healthy

Candidate NGB-20260907-G2.6 · NiceGUI Base 3.0.0a8 · NiceGUI 3.15.0

Generic application patterns

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Discoverable10 / 10

Live examples10 / 10

Semiconductor analytics

Available58

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Semiconductor recipes

Available8

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Catalog family coverage

Every capability family must be backed by at least one discoverable canonical authority entry.

shell/layout

1 discoverable entry

controls

35 discoverable entries

forms/overlays

tables



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REFERENCE

AI Development Guide

A compact, task-first guide for humans and AI agents using the installed NiceGUI Base APIs.

Search

⌘K / Ctrl+K

Theme · System

Appearance

Requirement → runnable reference

The same deterministic sequence is readable by a person and discoverable by an agent.

1 · REQUIREMENT

Investigate chamber drift and compare affected/control distributions

2 · RECOMMENDATION

Pattern: analysis_workspace · Recipe: excursion-defense-line

3 · AUTHORITIES

analytics:fdc_multi_sensor · analytics:rca_affected_control · component:data_table

4 · SCAFFOLD

nicegui-base create-recipe ./chamber-drift --name "Chamber Drift Review" --recipe excursion-defense-line

5 · BUSINESS/DOMAIN LOGIC

app/services and app/repositories; keep provider queries and domain calculations out of page callbacks

6 · EXTENSION RULE

Prefer registered pattern/component/visualization and token APIs; isolate a documented escape hatch only when no authority fits

7 · EXPECTED RESULT

A separately runnable app with governed filters, linked chart/table state, accessible empty/error states, and domain logic isolated in app-owned services

```
nicegui-base agent-context "Investigate chamber drift and compare affected/control distributions"
nicegui-base recommend-pattern "investigate chamber drift"
nicegui-base recommend-visualization "compare affected control distributions"
```



NiceGUI Base Reference Lab

Enterprise engineering application framework

LOCAL

Good morning
Process Engineer

PE

<

START

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Charts

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Engineering & RCA

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NiceGUI Base / Metrology Enginee...

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NiceGUI 3.15.0

LOCAL

Foundation

Inspect the exact visual primitives from which every application is constructed.

THEME

SystemLightDark

DENSITY

ComfortCompactDense

MOTION

NormalReduced

Resize the real browser window for responsive proof

Semantic color system

These are design roles, not a decorative palette. Apps should never invent independent visual colors.

Page

--cui-page

Primary surface

--cui-surface

Secondary surface

--cui-surface-secondary

Elevated

--cui-surface-elevated

Accent

--cui-accent

Success

--cui-success

Warning

--cui-warning

Danger

--cui-danger

Info

--cui-info

Primary text

--cui-text-primary

Secondary text

--cui-text-secondary

Border

--cui-border-default

Typography

Dense enterprise readability with restrained hierarchy and tabular numerics where values must scan precisely.

Type scale

Display — Strategic overview

Page — Equipment Health

Heading — Affected population

PNG SHA-256: 3d8b3b8753a2ae7b62cc83914f0c1f296a71c76d420970467b086775aed51ef7

page=13



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LOCAL

Shell Primitives

One responsive navigation state machine: expanded desktop, icon-only desktop, or temporary mobile overlay.

THEME

System

Light

Dark

DENSITY

Comfort

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Normal

Reduced

Resize the real browser window for responsive proof

Responsive navigation contract

Desktop has one collapse control inside the rail. Below 900px the rail is removed from interaction and one navigation trigger appears with the header actions — never beside the application title.

Expanded desktop

Icon-only desktop

Mobile overlay

Resize the browser to exercise the states. There is intentionally no standalone mobile-navigation demo action because that created a competing navigation system.

Header interaction contract

Application settings and user profile are real structured popovers. Settings navigates to the canonical settings workspace; the user control exposes identity and preferences instead of firing a toast-only action.



Structural acceptance rule

Exactly one navigation affordance is interactive per viewport. Sidebar/footer compression, duplicate menu triggers, content underlap, or lost navigation after resize are release blockers.



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Forms, Overlays & Feedback

Exercise real form anatomy, dirty state, dialogs, drawers, menus, progress, alerts and failure recovery.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

Production form

Routing

Changes here are intentionally marked dirty for navigation protection.

Preferred tool

ETCH-021



Recipe *

ETCH_R18



Automatic dispatch

Advanced limits



Collapsible form-section normalization.

Warning threshold

41.5

nm

Stop threshold

42.5

nm



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Enterprise DataTable Gallery

Five distinct production patterns prove different table jobs instead of repeating one measurement dataset.

THEME

System

Light

Dark

DENSITY

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Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

1 · Measurement population

High-volume read-only analysis: search, sort, filter, resize, select, conditional values, status, sparklines, bulk actions and row inspection.

Measurement population

320 deterministic records · dense analytical browsing · double-click any row for inspection

Q Search records				Columns		Compact		Export		Refresh	
☐	Measurement		Lot		Wafer		Tool		Parameter		Actions
☐	MEAS-100000		L260100		W01		ETCH-014		CD		... Actions
☐	MEAS-100001		L260100		W02		ETCH-021		Thickness		... Actions
☐	MEAS-100002		L260100		W03		CVD-008		Overlay		... Actions
☐	MEAS-100003		L260100		W04		CMP-004		Etch Rate		... Actions
☐	MEAS-100004		L260100		W05		PVD-011		CD		... Actions
☐	MEAS-100005		L260100		W06		MET-030		Thickness		... Actions
☐	MEAS-100006		L260100		W07		ETCH-014		Overlay		... Actions
☐	MEAS-100007		L260100		W08		ETCH-021		Etch Rate		... Actions
☐	MEAS-100008		L260101		W09		CVD-008		CD		... Actions
☐	MEAS-100009		L260101		W10		CMP-004		Thickness		... Actions
☐	MEAS-100010		L260101		W11		ETCH-014		Overlay		... Actions



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Visualization Recipe Gallery

Compact reusable recipes demonstrate analytical intent, markings, statistics, interactions and specialist visual forms—not just renderer types.

THEME

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Normal

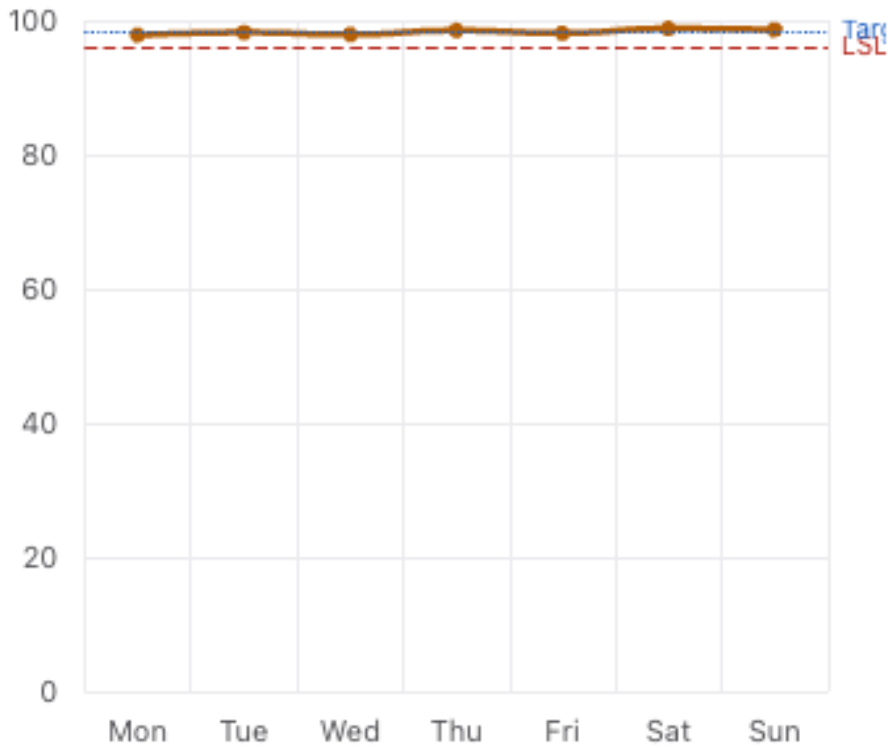
Reduced

Resize the real browser window for responsive proof

Trend & time recipes

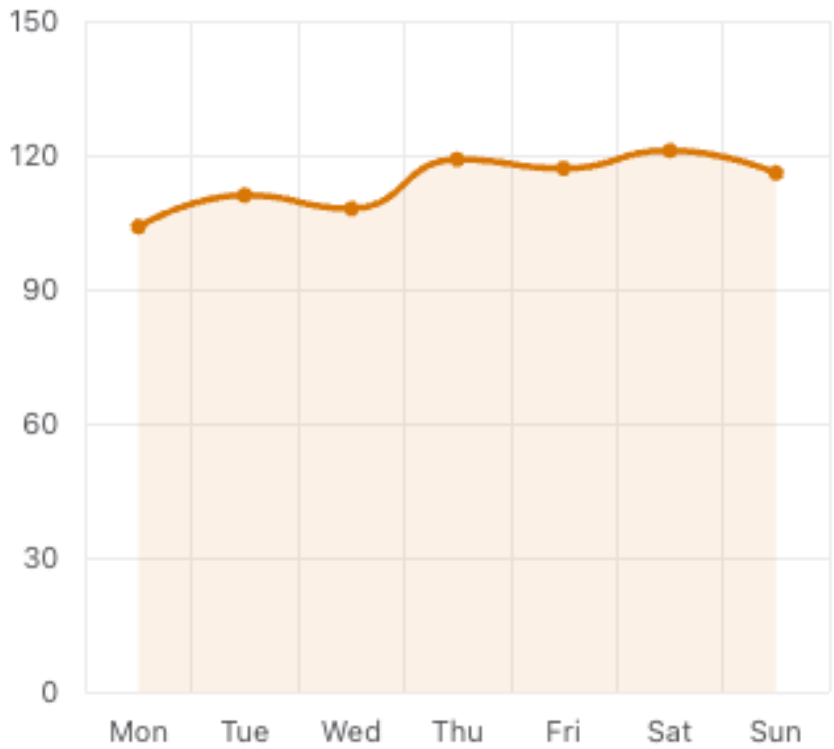
Premade temporal patterns include limits, target context, multi-series comparison, events, confidence bands and step changes.

Yield trend + target



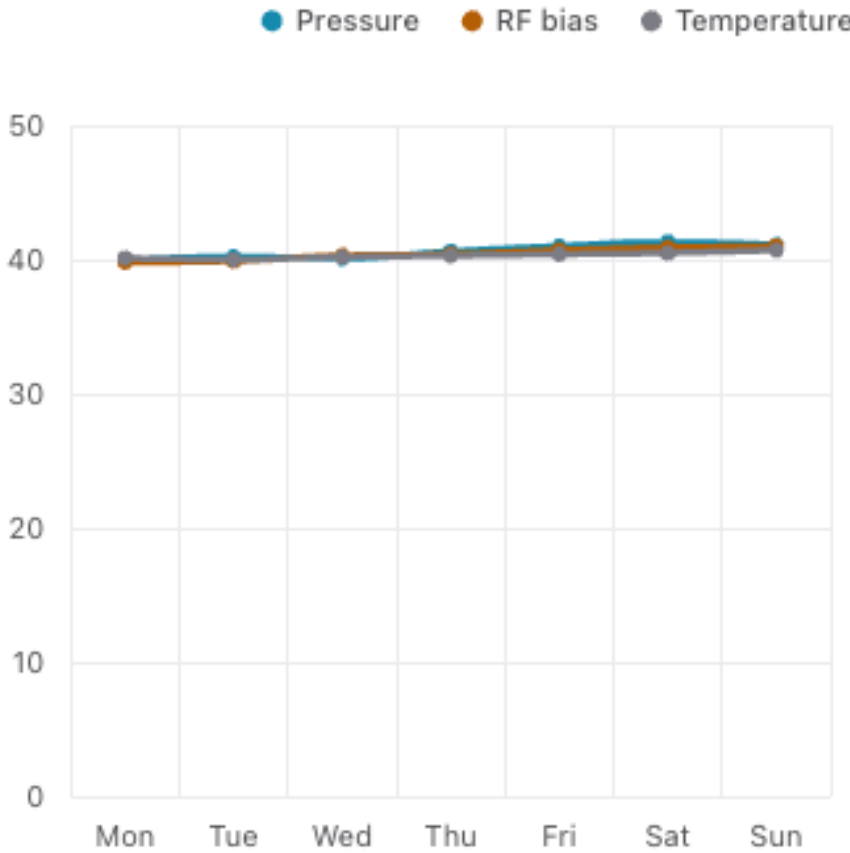
Show chart data

Throughput area



Show chart data

Multi-sensor overlay



Show chart data



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Content, Workflow & Commands

Viewers and workflow components should visually inherit NiceGUI Base rather than their underlying third-party defaults.

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Resize the real browser window for responsive proof

Entity and property presentation



ETCH-021 / CH-3

Watch

Critical etch chamber investigation

Chamber Area: ETCH Recipe: ETCH_R18

Key/value

Lot

L260142

Wafer

W12

Owner

Process Engineering

Property grid

STATUS

Watch

LAST PM

17 days

PRESSURE

12.8 mTorr

Comparison metric

Mean CD 42.18 nm 40.06 nm +2.12

Delta and trend indicators

Standalone metric indicators preserve non-color directional semantics.

↑ +2.12 nm ↓ -1.8% — Stable

Hierarchy and viewers

NiceGUI Base Reference Lab

Enterprise engineering application framework

LOCAL

Good morning
Process Engineer

PE

<

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LOCAL

Engineering & RCA Laboratory

Purpose-built investigation cockpit: scope, process behavior, population separation, evidence, commonality, contradictions and timeline.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

Affected population

84 wafers

12 lots

Matched controls

215 wafers

exact route strategy

Evidence confidence

High

2 strong channels

Contradictions

1

retained in decision

INVESTIGATION

EXC-1042 · Chamber 3 process degradation

OWNER

Process Engineering

STAGE

Evidence review

UPDATED

2 minutes ago

Investigation identity and specification context

ETCH-021 · Chamber 3

Etch critical dimension

Watch

TOOL / CHAMBER

ETCH-021 / CH-3

RECIPE

ETCH_R18

LAST PM

17 days

LOTS AFFECTED

12 lots

Specification semantics

Normal

Watch

Critical

40.100 nm · In spec

41.950 nm · Near upper limit

43.100 nm · Above USL

43.100 nm · Above USL

Current

42.180 nm

Baseline

40.060 nm

Delta

+2.120 nm

High

PNG SHA-256: baf9ce2f9a46aba6e416c218fea8fd3e6ee482b9aee92f3f642fb368339e0edf page=20



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NiceGUI Base / Metrology Enginee...

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NiceGUI Base 3.0.0a8
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State & Failure Laboratory

Every unhappy path must still look intentional and preserve clear recovery behavior.

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Resize the real browser window for responsive proof

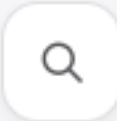
Canonical application states



No data yet

Connect a source or adjust the time window.

Configure



No matching results

Filters excluded all 18,402 measurements.

Clear filters



Unable to load this content

The synthetic service returned a controlled failure.

Error ID: ERR-LAB-2048

Retry



Connection unavailable

The connection is unavailable. Existing content should remain understandable.

Retry



Access restricted

You do not have permission to view this investigation.



Page not found

The requested synthetic investigation does not exist.

Go back



Generic state view



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Performance Laboratory

Interactive workloads expose generation, render assembly, cancellation and query timing instead of hiding work behind a button.

THEME

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MOTION

Normal

Reduced

Resize the real browser window for responsive proof

Interactive workload benchmark

Every action updates visible state immediately. The large workload remains opt-in so ordinary lab navigation stays fast.

Local grid workload

Generate deterministic synthetic rows off the UI event loop, mount the real Company DataTable, then expose timing and status.

Idle

0

Rows

—

Generation

—

Table mount

—

Filter benchmark



No stress workload loaded

Choose 10k first, then 100k when you want the full local stress path.

Load 10k rows

Load 100k rows

Cancel / clear

Background-work presentation

Company Engineering Workspace

Reference applications · synthetic engineering data

Good morning
Process Engineer

PE

<

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REFERENCE APP PREVIEW · DASHBOARD

Dashboard

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Yield Performance

Executive-to-engineering overview with one clear reading order

Area

All areas × ▾

24 h

7 d

30 d

🕒 Yield

98.4%

↑ +0.7% WoW

📉 Excursions

12

↑ 3 critical

📅 Fleet health

94.6%

— stable

🕒 Stale signals

4

🕒 >30 min

Yield trend

+ − ⌵ ↺ 📅 ⬇️ ⌵

100

80

60

40

Excursio
n state

+ − ⌵ ↺ 📅 ⬇️ ⌵

● Resolved

● Monitoring

● Open

PNG SHA-256: 5b26c09de89c10deb5625ef93381c3725ca24f7100edb5594b6411082fdc1b75

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REFERENCE APP PREVIEW · DATA EXPLORER

Data Explorer

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Process Measurement Explorer

Filter → summarize → analyze → inspect without losing population context

Area

Tool

All areas



All tools



24 h

7 d

30 d

Population

18,402

selected rows

Mean CD

40.62 nm

+0.31 nm

OOS

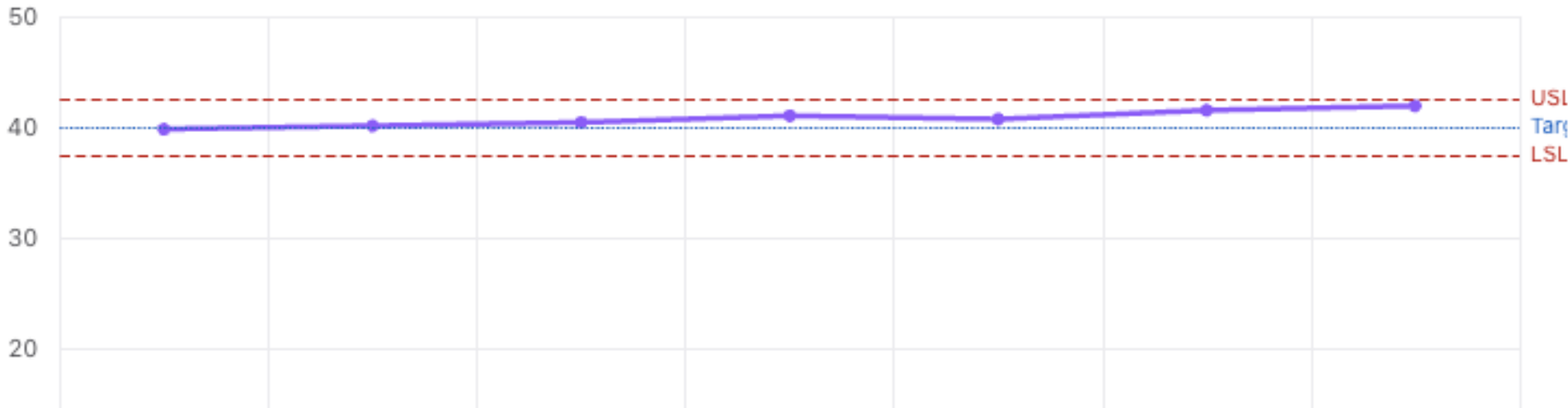
2.8%

516 wafers

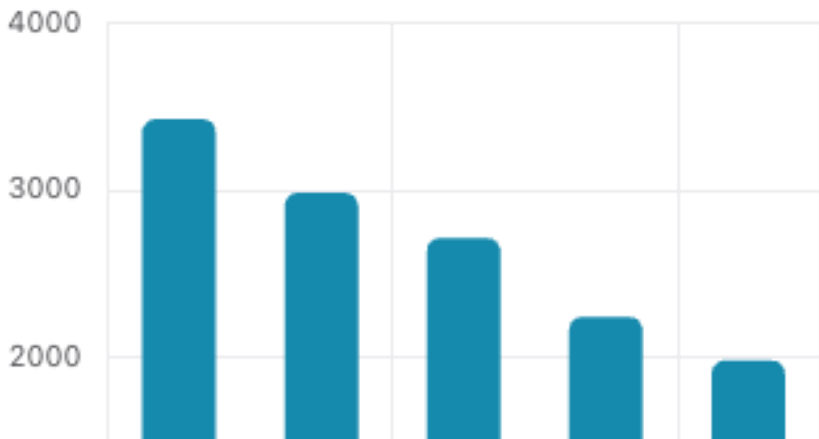
Freshness

4 min

Selected parameter



Populati
on by
tool



Company Engineering Workspace

Reference applications · synthetic engineering data

Good morning

Process Engineer

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REFERENCE APP PREVIEW · MASTER DETAIL

Master Detail

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Equipment Fleet

Browse a dense fleet while selected-equipment context stays visible

Area

Tool

All areas × ▾

All tools × ▾

Equipment signals

Search records /

Columns

Compact

Export

Refresh

	Measurement ▾	Lot ▾	Wafer ▾
☐	MEAS-100000	L260100	W01
☐	MEAS-100001	L260100	W02
☐	MEAS-100002	L260100	W03
☐	MEAS-100003	L260100	W04
☐	MEAS-100004	L260100	W05
☐	MEAS-100005	L260100	W06
☐	MEAS-100006	L260100	W07
☐	MEAS-100007	L260100	W08

ETCH-021 / CH-3

Watch

Etch chamber

RECIPE

ETCH_R18

LAST PM

17 days

HEALTH

82%

Recent CD

+ -

50

PNG SHA-256: c23d8e8f2b003a1cfa837c438d9f9072f2b65437ed5f5be2b41016a0f76ecd0ed page=25

Crud

interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

[Back to Reference Explorer](#)

Saved Investigation Views

Create, inspect and manage reusable engineering configurations

Search saved views

All

Mine

Shared

+ New saved view

Saved views

Q Search records

/

Columns

≡ Compact

 Export

Refresh

ID	Name	Owner	Status
☐ v1	Critical chambers	You	● Active
☐ v2	Daily yield review	You	● Active
☐ v3	PM correlation	Process team	● Shared

Page Size: 50

1 to 3 of 3

Page 1 of 1

3 records

Compact · 38 px

Company Engineering Workspace

Reference applications · synthetic engineering data

Good morning
Process Engineer

PE

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REFERENCE APP PREVIEW · SEARCH

Search

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Engineering Search

Search-first navigation for tools, chambers, investigations and evidence

Search everything

ETCH-021

Refine

Equipment

Investigations

Evidence

3 results · indexed 2 min ago

ETCH-021

Tool · Watch

3 active excursions · 2 evidence packages

ETCH-021 / CH-3

Chamber · Watch

Strong commonality with EXC-1042

EXC-1042

Investigation · Active

CD drift and edge spatial signature

PNG SHA-256: c9c842f55e2e682038cd3a64e6f4d8ef3fa4ae3cc7f9b7b25494beb3928a6875 page=28

Company Engineering Workspace

Reference applications · synthetic engineering data

Good morning
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REFERENCE APP PREVIEW · WIZARD

Wizard

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Create Investigation

Guided task with visible progress, bounded decisions and a real review step

Scope

Complete

Controls

Active

3

Review

Upcoming

Select control population

Use the narrowest defensible comparison population before reviewing evidence.

Control strategy

Exact route match

Require exact tool / recipe / chamber sequence

Exclude wafers processed after maintenance

Cancel

Review controls

PNG SHA-256: 4bdee1b2e2c52cf07ceca811c528f9bb997105c9ec74d6982dc7927dd4f0f0b5 page=29

Company Engineering Workspace

Reference applications · synthetic engineering data

☀️

Good morning

Process Engineer

PE

<

START

🏠

Overview

🧩

Foundation

📄

Shell Primitives

ELEMENTS

☀️

Controls

✏️

Forms & Overlays

📊

DataTable Lab

📈

Charts

📄

Content & Workflow

🌐

Engineering & RCA

REFERENCE APPS

🏠

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🔍

Data Explorer

📁

Master / Detail

👤

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REFERENCE APP PREVIEW · COMPARISON

Comparison

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Affected vs Control

Direct comparison keeps population identity, deltas and spatial evidence aligned

Affected population

EXC-1042 · 84 wafers

×

▼

Control population

Matched route · 215 wafers

×

▼

Affected mean

42.18 nm

Control mean

39.96 nm

Delta

+2.22 nm

SMD

1.42

AFFECTED

📄

ETCH-021 / CH-3

Affected

84 wafers · excursion population

CONTROL

📄

ETCH-014 / CH-2

Control

215 wafers · matched route

Spatial signature

Affected edge signature vs matched control on one scale

+

−

↺

AFFECTED

97 measured dies

CONTROL

97 measured dies

Shared scale

PNG SHA-256: a09300454f62489907563d7fd0d9fbc1e4909eaa144d444a0b39239dbe1d91ae page=30

<

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

⚙️

Theme · System

Appearance

COMPONENT · ACTIONS

ButtonGroup

Group related actions without visual fragmentation

BEST FOR

toolbar actions · mode actions

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

>

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

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ButtonGroup

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Run

Pause

Reset

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Theme · System

Appearance

COMPONENT · ACTIONS

SplitButton

Primary action with closely related alternatives

BEST FOR

export alternatives · run options

AVOID WHEN

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> **Advanced contract · configuration, data, accessibility, and authority**

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PreviewUsageInspectCode

SplitButton

Interactive specimen Use the real control below; interactions are recorded in Event / debug output.

Export CSV

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Department standard · Reference Explorer

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Theme · System

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COMPONENT · SURFACES

Divider

Subtle structural separation

BEST FOR

sections · menus

AVOID WHEN

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Advanced contract · configuration, data, accessibility, and authority

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Live example is ready.

Preview

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Divider

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Measurement summary

REFERENCE

Member Member

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · SURFACES

CollapsiblePanel

Progressive disclosure without leaving context

BEST FOR

advanced settings · secondary detail

AVOID WHEN

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Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

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CollapsiblePanel

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Advanced settings

PNG SHA-256: 330b59e920c68f80cb1fe28e05ec0e80114ecf5ef6370ce127a70731f9442c1d page=35

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Search

⌘K / Ctrl+K



Theme · System

Appearance

COMPONENT · SURFACES

Accordion

Grouped progressive disclosure

BEST FOR

help · configuration groups

AVOID WHEN

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› **Advanced contract · configuration, data, accessibility, and authority**

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Accordion

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Sampling



Every 10 wafers

<

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⚙️

Theme · System

Appearance

COMPONENT · STATUS

Chip

Interactive compact metadata or filter value

BEST FOR
filter value · selection

AVOID WHEN
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› **Advanced contract · configuration, data, accessibility, and authority**

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Preview Usage Inspect Code

Chip

Interactive specimen

 Use the real control below; interactions are recorded in Event / debug output.

📦

 Lot A12

REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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Theme · System

Appearance

COMPONENT · STATUS

CountBadge

Compact numeric count

BEST FOR

notifications · selected count

AVOID WHEN

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> Advanced contract · configuration, data, accessibility, and authority

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CountBadge

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

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Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · STATUS

SeverityIndicator

Semantic severity with label and non-color cue

BEST FOR

operational state · risk

AVOID WHEN

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> Advanced contract · configuration, data, accessibility, and authority

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Live example is ready.

Preview

Usage

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SeverityIndicator

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

● High severity

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · STATUS

FreshnessIndicator

Data recency state

BEST FOR

updated time · stale data

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

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FreshnessIndicator

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Updated 2 min ago

REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · STATUS

DataQualityBadge

Data reliability/completeness state

BEST FOR

partial data · estimated data

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

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DataQualityBadge

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Complete

NiceGUI Base

Department standard · Reference Explorer

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · ACTIONS

Button

Standard user action

BEST FOR

normal action · secondary action

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

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Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Run analysis

REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · ACTIONS

ActionButton

Async-safe or stateful action

BEST FOR

save · run analysis

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

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ActionButton

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Save changes

REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · ACTIONS

IconButton

Compact icon-only action with accessible label

BEST FOR

toolbar action · row action

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

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Code

IconButton

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · SURFACES

Surface

Consistent content containment and interaction surface

BEST FOR

panel · card

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

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Surface

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Panel

Card

Well

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K



Theme · System

Appearance

COMPONENT · STATUS

Badge

Compact semantic status or metadata label

BEST FOR

status · severity

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

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Badge

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.



Healthy



Warning



Excursion

DISCOVER

🏠

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NiceGUI Base 3.0.0a8

NiceGUI 3.15.0

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

⚙️ Theme · System

Appearance

COMPONENT · INPUTS

TextInput

Single-line text entry

BEST FOR

name · identifier

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview Usage Configure States Interactions Inspect Code

TextInput

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Lot ID

LOT-2409

×

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⚙️

Theme · System

Appearance

COMPONENT · INPUTS

NumberInput

Numeric entry with bounds and unit

BEST FOR

threshold · count

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

>

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

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NumberInput

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Upper limit

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K



Theme · System

Appearance

COMPONENT · INPUTS

TextArea

Multi-line text entry

BEST FOR

notes · description

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

TextArea

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Engineering note

Investigate chamber drift.

REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

SearchInput

Debounced search entry

BEST FOR

global search · table search

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Configure

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Interactions

Inspect

Code

SearchInput

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Search lots

REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

Select

Single choice from known values

BEST FOR

filter · form choice

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

Select

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Process area

REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

MultiSelect

Multiple choices from known values

BEST FOR

multi-filter · assignment

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Configure

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Interactions

Inspect

Code

MultiSelect

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Process areas

REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

Autocomplete

Searchable known values

BEST FOR

large option set · entity lookup

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

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Autocomplete

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

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Search

⌘K / Ctrl+K



Theme · System

Appearance

COMPONENT · INPUTS

Combobox

Search/select with optional custom value

BEST FOR

tag-like entry · mixed known/custom choice

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

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Combobox

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Tag

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K



Theme · System

Appearance

COMPONENT · INPUTS

Checkbox

Independent boolean selection

BEST FOR

multi-option form · enable choice

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

Checkbox

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.



Include rework lots

REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

CheckboxGroup

Set of independent boolean choices

BEST FOR

feature selection · permissions

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

CheckboxGroup

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Signals

☒ Etch

☐ Deposition

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

RadioGroup

Mutually exclusive choice

BEST FOR

mode selection · single preference

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

RadioGroup

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Mode

Etch

Deposition

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

Switch

Immediate on/off setting

BEST FOR

preference toggle · feature enablement

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

Switch

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Auto refresh

REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

Slider

Bounded continuous or stepped value

BEST FOR

threshold · range tuning

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

Slider

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Threshold

65%

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K



Theme · System

Appearance

COMPONENT · INPUTS

RangeSlider

Bounded low/high selection

BEST FOR

numeric filtering · window selection

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

RangeSlider

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Window

20–80%

REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

DatePicker

Single date selection

BEST FOR

effective date · event date

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

DatePicker

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Effective date

REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

DateRangePicker

Date interval selection

BEST FOR

analysis period · reporting window

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

DateRangePicker

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Analysis period

PNG SHA-256: ff3501ec0d58e082e6bb98fd2c682ad32ac1cfc6e27647aedf85f57efe4baa90 page=62



Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K



Theme · System

Appearance

COMPONENT · INPUTS

TimePicker

Time-of-day selection

BEST FOR

schedule time · cutoff

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

› Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Configure

States

Interactions

Inspect

Code

TimePicker

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Cutoff time

REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

DateTimePicker

Date and time selection

BEST FOR

timestamp · scheduled action

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Configure

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Inspect

Code

DateTimePicker

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Scheduled run

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

COMPONENT · INPUTS

FileUpload

Validated file selection and upload

BEST FOR

attachment · data import

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Configure

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Interactions

Inspect

Code

FileUpload

Interactive specimen

Use the real control below; interactions are recorded in Event / debug output.

Upload process data

Click to browse, drag files here, or paste from the clipboard with Ctrl+V / Cmd+V.

0.00B / 0.00%

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search

⌘K / Ctrl+K



Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Dashboard

High-level KPI and trend overview.

BEST FOR

High-level KPI and trend overview.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Data

Inspect

Code

Dashboard

At-a-glance process performance with trends and exceptions.

Lots today

128

In control

96.9%

Open exceptions

2

REFERENCE

Member

Member

M

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

⚙️

Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Data Explorer

Interactive filtering, analysis, records and contextual drill-down.

BEST FOR

Interactive filtering, analysis, records and contextual drill-down.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

›

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Data

Inspect

Code

Data Explorer

Filter records, inspect the governed table, and retain selected context.

Search lots

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Master Detail

Browse entities while preserving selected-entity context.

BEST FOR

Browse entities while preserving selected-entity context.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Data

Inspect

Code

Master Detail

Browse lots while the selected lot remains visible beside the master list.

Find lot or tool

ETCH

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REFERENCE

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Crud

Search, create, inspect and edit managed records.

BEST FOR

Search, create, inspect and edit managed records.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

PreviewUsageDataInspectCode

Crud

Create, read, update, validate, and safely delete managed records.

Search managed recipes

POLY

Create recipeDelete selected...

<

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

⚙️

Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Monitoring

Operational health, alerts and periodically refreshed data.

BEST FOR

Operational health, alerts and periodically refreshed data.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

>

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

Data

Inspect

Code

Monitoring

Lead with operational health, trend, alerts, and supporting evidence.

Fleet health

11 / 12

Active alarms

2

Data freshness

18 s

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search⌘K / Ctrl+K

Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Search

Search and refine heterogeneous or entity-oriented results.

BEST FOR

Search and refine heterogeneous or entity-oriented results.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

PreviewUsageInspectCode

Search

Query and refine entity results without making an empty state the default.

Search lots, tools, or recipes

ETCH

8 results for "ETCH" · Status: Any status

LOT-240901

ETCH 02 · Chamber A

<

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Search ⌘K / Ctrl+K

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Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Settings

Structured application/user configuration.

BEST FOR

Structured application/user configuration.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

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Settings

Navigate settings, edit governed fields, validate, save, and reset local state.

Settings sections

Data refresh

Data refresh

These values apply to this local specimen only.

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Theme · System

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PATTERN · APPLICATION PATTERN

Wizard

Guided multi-step task with clear progress and bounded decisions.

BEST FOR

Guided multi-step task with clear progress and bounded decisions.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

> **Advanced contract · configuration, data, accessibility, and authority**

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PreviewUsageInspectCode

Wizard

Complete a bounded task with progress, validation, review, Back, and Next.



REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Comparison

Compare baseline/current populations, entities or scenarios.

BEST FOR

Compare baseline/current populations, entities or scenarios.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

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Comparison

Align two populations and make deltas and evidence immediately comparable.

Baseline

Golden tool · ETCH-03

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Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search ⌘K / Ctrl+K

Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Analysis Workspace

Maximum-density resizable chart/table analysis environment.

BEST FOR

Maximum-density resizable chart/table analysis environment.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Data

Inspect

Code

Analysis Workspace

Keep analysis controls, visualization, records, and selected context together.

Tool

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Department standard · Reference Explorer

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Search ⌘K / Ctrl+K

Theme · System

Appearance

ANALYTIC · SPC

SPC I-MR

Individuals and moving-range control chart

BEST FOR

continuous measurements without rational subgroups

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

SPCMonitor

PMEffectAnalysis

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

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Control chart · Individual + moving range · x: sample order · y: measurement

Individuals + Moving Range · x: sample order / adjacent samples · y: Measurement / absolute difference

Individuals — SPC I-MR

+ - [] ↺ [] ⌵ []



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ANALYTIC · SPC

SPC Xbar-R

Xbar-R subgroup control chart

BEST FOR

small equal-size rational subgroups

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

› **Advanced contract · configuration, data, accessibility, and authority**

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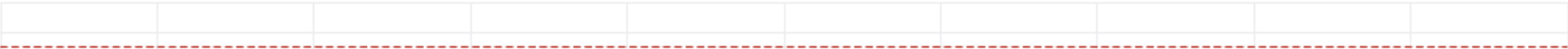
Canonical synthetic reference fixture; engineering conclusions require validated production data.

X — SPC Xbar-R



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UCI



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Theme · System

Appearance

ANALYTIC · SPC

SPC P

Proportion nonconforming control chart

BEST FOR

binary defect proportion with varying samples

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

> **Advanced contract · configuration, data, accessibility, and authority**

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

SPC P



● Nonconforming proportion ● Center ● UCL ● LCL

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Theme · System

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ANALYTIC · SPC

SPC Np

Count nonconforming control chart

BEST FOR
binary defect counts with constant sample size

AVOID WHEN
when chart assumptions, subgroup structure, or baseline definition do not match the data

> **Advanced contract · configuration, data, accessibility, and authority**

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

SPC Np

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● Nonconforming count ● Center ● UCL ● LCL

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⚙️

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ANALYTIC · SPC

SPC U

Defects-per-unit control chart

BEST FOR

Poisson defects with varying opportunity

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

>

Advanced contract · configuration, data, accessibility, and authority

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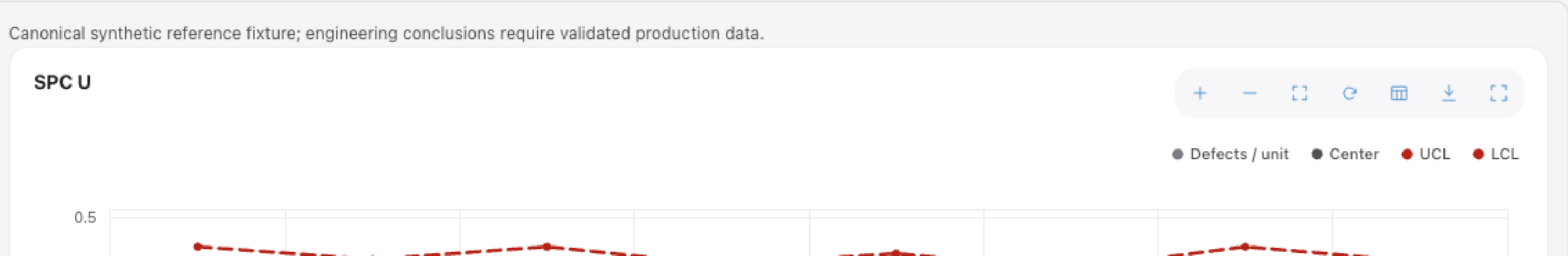
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☼ Theme · System

Appearance

ANALYTIC • SPC

SPC CUSUM

CUSUM change-detection chart

BEST FOR

small persistent mean shifts

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

- › **Advanced contract · configuration, data, accessibility, and authority**

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Control chart · cumulative sum · x: sample order · y: cumulative deviation

SPC CUSUM

● C+ ● C- ● Center ● Positive decision limit ● Negative decision limit



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Theme · System

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ANALYTIC · CAPABILITY

Capability Histogram

Capability distribution with spec limits

BEST FOR

Cp/Cpk/Pp/Ppk and distribution shape

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

SPCMonitor

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Histogram · x: measurement bins · y: Count

Capability Histogram

Numeric measurement bins with specification markers on the same coordinate axis.

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ANALYTIC · CAPABILITY

Q-Q Probability

Normal probability/QQ diagnostic

BEST FOR

capability normality assumption review

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

› **Advanced contract · configuration, data, accessibility, and authority**

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Code

Q-Q probability plot · x: Theoretical quantile · y: Observed value

Q-Q Probability

Observed points against a fitted location/scale normal reference line.

Observed Expected reference



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Appearance

ANALYTIC · CAPABILITY

ECDF

Empirical cumulative distribution

BEST FOR

distribution comparison without binning

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

› **Advanced contract · configuration, data, accessibility, and authority**

Related capabilities

SPCMonitor

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Preview

Usage

Data

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Code

Empirical cumulative distribution · x: Value · y: Cumulative probability

ECDF



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Appearance

ANALYTIC · CAPABILITY

Box Distribution

Box distribution comparison

BEST FOR

compact affected/control distribution summaries

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

› **Advanced contract · configuration, data, accessibility, and authority**

Related capabilities

PMEffectAnalysis

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

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Code

Box distribution · x: population · y: measurement

Box Distribution

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Theme · System

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ANALYTIC · CAPABILITY

Violin Distribution

Violin distribution comparison

BEST FOR

shape plus spread comparison

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

Advanced contract · configuration, data, accessibility, and authority

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Live example is ready.

Preview

Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Violin distribution · x: population · width: local density

Violin Distribution

Density is estimated from the canonical population measurements.

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Theme · System

Appearance

ANALYTIC · CAPABILITY

Ridge Distribution

Ridge distribution comparison

BEST FOR

many tool/chamber distributions

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

› Advanced contract · configuration, data, accessibility, and authority

Related capabilities

ChamberMatching

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Preview

Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Ridge distribution · rows: populations · x: value · height: density

Ridge Distribution

Offset density curves keep each chamber distribution readable.



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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Continuous

Continuous wafer map

BEST FOR

continuous spatial metrology or process response

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

› Advanced contract · configuration, data, accessibility, and authority

Related capabilities

SPCMonitor

LotWaferExplorer

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Preview

Usage

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Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Continuous

Die-level wafer signature · wheel/drag to inspect spatial structure



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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Categorical

Categorical/bin wafer map

BEST FOR

yield bins or categorical spatial state

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

[LotWaferExplorer](#)

[YieldLoss](#)

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Preview

Usage

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Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Categorical wafer map · legend: Category · die position: x/y

Wafer Categorical

Die-level wafer signature · wheel/drag to inspect spatial structure



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Theme · System

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ANALYTIC · WAFER

Wafer Defect

Defect wafer map

BEST FOR

inspection defects and defect classes

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract · configuration, data, accessibility, and authority

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Preview

Usage

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Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Defect wafer map · legend: Defect state · die position: x/y

Wafer Defect

Die-level wafer signature · wheel/drag to inspect spatial structure



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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Delta

Affected minus control wafer delta map

BEST FOR

spatial process change localization

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

› Advanced contract · configuration, data, accessibility, and authority

Related capabilities

ExcursionDefenseLine

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Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Delta

Die-level wafer signature · wheel/drag to inspect spatial structure



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Theme · System

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ANALYTIC · WAFER

Wafer Comparison

Side-by-side wafer comparison

BEST FOR

affected/control wafer signature comparison

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

ChamberMatching

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Live example is ready.

Preview

Usage

Data

Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Comparison

Synchronized affected vs control wafer signature · one shared scale

PNG SHA-256: ffd0e262b25efaed97e4bb3952c3973f17e4ed25c96aa29c4293622f3394ffd page=95



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Theme · System

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ANALYTIC · WAFER

Lot Wafer Strip

Lot wafer strip

BEST FOR

lot-level wafer-to-wafer spatial variation

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

› Advanced contract · configuration, data, accessibility, and authority

Related capabilities

LotWaferExplorer

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Preview

Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Lot Wafer Strip

Wafer W01

Die-level wafer



Wafer W02

Die-level wafer



Wafer W03

Die-level wafer



Wafer W04

Die-level wafer



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ANALYTIC · WAFER

Wafer Small Multiples

Synchronized wafer small multiples

BEST FOR

many wafers on one scale

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract · configuration, data, accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Data

Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Small Multiples

Wafer W01

Die-level wafer signature · wheel/drag to inspect spatial structure

Wafer W02

Die-level wafer signature · wheel/drag to inspect spatial structure

Wafer W03

Die-level wafer signature · wheel/drag to inspect spatial structure

Wafer W04

Die-level wafer signature · wheel/drag to inspect spatial structure

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Theme · System

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ANALYTIC · WAFER

Wafer Contour

Contour/surface wafer map

BEST FOR

data-derived spatial field interpretation

AVOID WHEN

sparse categorical data

› **Advanced contract · configuration, data, accessibility, and authority**

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Preview

Usage

Data

Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Contour

Data-derived continuous wafer field · isolines are clipped to governed wafer geometry



CLIPPED DATA-DERIVED CONTOUR ISOLINES



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ANALYTIC · WAFER

Wafer Radial

Radial profile

BEST FOR

center-edge signatures

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

› **Advanced contract · configuration, data, accessibility, and authority**

Related capabilities

LotWaferExplorer

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Preview

Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Radial

Center → edge profile · compare radial signature without Cartesian chart chrome

REFERENCE

Member Member

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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Theme · System

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ANALYTIC · WAFER

Wafer Center Edge

Center-edge decomposition

BEST FOR

quantify center versus edge shift

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract · configuration, data, accessibility, and authority

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Center Edge

+ - [] ↺ [] ⌵ []

50 nm

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ANALYTIC · WAFER

Wafer Ring

Ring analysis

BEST FOR

concentric process signatures

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

LotWaferExplorer

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Ring

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ANALYTIC · WAFER

Wafer Sector

Sector analysis

BEST FOR

azimuthal spatial signatures

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

› **Advanced contract · configuration, data, accessibility, and authority**

Related capabilities

LotWaferExplorer

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Sector



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ANALYTIC · WAFER

Wafer Defect Clusters

Defect-cluster overlays

BEST FOR

identify spatially clustered inspection defects

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

LotWaferExplorer

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Wafer Defect Clusters

Die-level wafer signature · wheel/drag to inspect spatial structure



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ANALYTIC · FDC

FDC Recipe Step Trace

Recipe-step-aligned trace

BEST FOR

compare traces with step semantics

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FdcToolHealth

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Preview

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Recipe Step Trace



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Theme · System

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ANALYTIC · FDC

FDC Golden Envelope

Golden trace envelope

BEST FOR

compare trace deviation against baseline population

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FdcToolHealth

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Golden Envelope



Golden upper

Observed

Golden lower



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Appearance

ANALYTIC · FDC

FDC Multi Sensor

Multi-sensor trace panel

BEST FOR

correlated equipment sensor behavior

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

› Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FdcToolHealth

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Multi Sensor

Pressure RF bias Gas flow

Reference Explorer

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Theme · System

Appearance

ANALYTIC · FDC

FDC Tool Chamber Compare

Tool/chamber comparison

BEST FOR

matching and PM-effect comparisons

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

› **Advanced contract · configuration, data, accessibility, and authority**

Related capabilities

[PMEffectAnalysis](#)

[ChamberMatching](#)

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Tool Chamber Compare



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ANALYTIC · FDC

FDC Chamber Fingerprint

Chamber fingerprint

BEST FOR

multivariate chamber matching

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FdcToolHealth

PMEffectAnalysis

ChamberMatching

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Chamber Fingerprint

Normalized chamber/process fingerprint · hover cells for exact contribution



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ANALYTIC · FDC

FDC Sensor Fingerprint

Sensor fingerprint

BEST FOR

sensor signature comparison

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

› **Advanced contract · configuration, data, accessibility, and authority**

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Preview

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Sensor Fingerprint

Normalized chamber/process fingerprint · hover cells for exact contribution



Mean shift

Slope

Noise



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ANALYTIC · FDC

FDC Alarm Overlay

Alarm overlay

BEST FOR

align alarms with trace excursions

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

› **Advanced contract · configuration, data, accessibility, and authority**

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Preview Usage Data Inspect Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Alarm Overlay

2.5

Pressure high



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Theme · System

Appearance

ANALYTIC · FDC

FDC Equipment Event Overlay

Equipment event overlay

BEST FOR

align PM/recipe/equipment events with traces

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

› Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FdcToolHealth

PMEffectAnalysis

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FDC Equipment Event Overlay





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Theme · System

Appearance

ANALYTIC · FDC

FDC PCA Scores

PCA scores

BEST FOR

multivariate operating-state separation

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

› Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FdcToolHealth

ChamberMatching

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC PCA Scores



● Baseline

● Affected

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ANALYTIC · FDC

FDC PCA Loadings

PCA loadings

BEST FOR

variables driving multivariate separation

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FdcToolHealth

ChamberMatching

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC PCA Loadings

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Theme · System

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ANALYTIC · FDC

FDC Hotelling T²

Hotelling T²

BEST FOR

multivariate score-space anomaly magnitude

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

› **Advanced contract · configuration, data, accessibility, and authority**

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PreviewUsageDataInspectCode

Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC Hotelling T²

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Theme · System

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ANALYTIC · FDC

FDC SPE Q

SPE/Q statistic

BEST FOR

PCA residual anomaly magnitude

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

>

Advanced contract · configuration, data, accessibility, and authority

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

FDC SPE Q

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Theme · System

Appearance

ANALYTIC · RCA

RCA Affected Control

Affected/control population comparison

BEST FOR

quantify excursion shift without causal overclaim

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

ExcursionDefenseLine

PMEffectAnalysis

RcaCockpit

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Live example is ready.

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Affected Control

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Theme · System

Appearance

ANALYTIC · RCA

RCA Commonality Ranking

Commonality ranking

BEST FOR

rank shared factors across affected versus control

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

ExcursionDefenseLine

YieldLoss

RcaCockpit

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Commonality Ranking

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Theme · System

Appearance

ANALYTIC · RCA

RCA Enrichment

Enrichment analysis

BEST FOR

factor exposure over-representation

AVOID WHEN

as proof of causality without corroborating engineering evidence

› **Advanced contract · configuration, data, accessibility, and authority**

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RCA Enrichment

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Appearance

ANALYTIC · RCA

RCA Commonality Matrix

Commonality matrix

BEST FOR

scan factor/value enrichment

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

ExcursionDefenseLine

RcaCockpit

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Commonality Matrix

RCA factor commonality · darker cells indicate stronger population enrichment

+ - ↺

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Theme · System

Appearance

ANALYTIC · RCA

RCA Contribution Waterfall

Contribution waterfall

BEST FOR

show ranked contribution accumulation

AVOID WHEN

as proof of causality without corroborating engineering evidence

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

RcaCockpit

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Live example is ready.

Preview

Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Contribution Waterfall

Signed bars span each contribution from its prior cumulative total to the reconciled net shift.

Cumulative delta

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Theme · System

Appearance

ANALYTIC · RCA

RCA Correlation Matrix

Correlation matrix

BEST FOR

numeric relationship screening

AVOID WHEN

causal conclusions

Advanced contract · configuration, data, accessibility, and authority

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Live example is ready.

Preview

Usage

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Correlation Matrix

+ - [] ↺ [] ⌵ ⌶

Temperature

PNG SHA-256: afe12d79e8d62370011a7a06139d192fef48892d4a495cb77b113c97037c3a6 page=121

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Theme · System

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ANALYTIC · RCA

RCA Evidence Matrix

Evidence matrix

BEST FOR

hypothesis evidence/contradiction review

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

ExcursionDefenseLine

RcaCockpit

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Live example is ready.

Preview

Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Evidence Matrix

RCA factor commonality · darker cells indicate stronger population enrichment

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ANALYTIC · RCA

RCA Genealogy Graph

Genealogy/process graph

BEST FOR

lot/wafer/tool/recipe route genealogy

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

ExcursionDefenseLine

RcaCockpit

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Live example is ready.

Preview

Usage

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Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Genealogy Graph

Branching and merging entity relationships preserve route order; they do not imply causality.



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Theme · System

Appearance

ANALYTIC · RCA

RCA Cause Tree

Cause tree

BEST FOR

structured causal candidates

AVOID WHEN

as proof of causality without corroborating engineering evidence

› Advanced contract · configuration, data, accessibility, and authority

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Preview

Usage

Data

Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Cause Tree

Hierarchical candidate decomposition; candidates remain hypotheses until evidence corroborates them.

CH-3 drift

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ANALYTIC · RCA

RCA Fault Tree

Fault tree

BEST FOR

logical failure decomposition

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

RcaCockpit

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Fault Tree

AND/OR logic is explicit and remains separate from evidence confidence.

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Theme · System

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ANALYTIC · RCA

RCA Sankey

Sankey/process flow

BEST FOR

population flow through route/tool/process paths

AVOID WHEN

as proof of causality without corroborating engineering evidence

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

[RcaCockpit](#)

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

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Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

RCA Sankey

Band width encodes wafer quantity through route, tool, chamber, and review stages.





Reference Explorer

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Theme · System

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ANALYTIC · YIELD

Yield Pareto

Yield loss Pareto

BEST FOR

rank yield-loss contributors

AVOID WHEN

when numerator/denominator or bin definitions are inconsistent across populations

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

YieldLoss

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Preview

Usage

Data

Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Yield Pareto

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Contributors Cumulative % Cumulative

+ / cumulative percent

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Theme · System

Appearance

ANALYTIC · YIELD

Bin Pareto

Bin Pareto

BEST FOR

rank failing bins

AVOID WHEN

when numerator/denominator or bin definitions are inconsistent across populations

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

YieldLoss

Reference example · governed sample data stays on this page and never modifies a project. Live example is ready.

Preview

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Inspect

Code

Canonical synthetic reference fixture; engineering conclusions require validated production data.

Bin Pareto

+ / cumulative percent

Contributors

Cumulative %

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ANALYTIC · YIELD

Yield Waterfall

Yield waterfall/decomposition

BEST FOR

explain yield delta composition

AVOID WHEN

when numerator/denominator or bin definitions are inconsistent across populations

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

YieldLoss

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

Yield Waterfall

Loss and recovery bars reconcile the signed total yield change.

Cumulative delta



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ANALYTIC · RELIABILITY

Weibull Reliability

Weibull reliability analysis

BEST FOR

time-to-failure reliability data

AVOID WHEN

when failure/censoring semantics or exposure time are unknown

› Advanced contract · configuration, data, accessibility, and authority

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

Weibull Reliability

β 1.35 · η 91.8 · R^2 0.96 · failures 9 · censored 3



Failures



Censored



Weibull fit

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ANALYTIC · DOE

DOE Main Effects

DOE main effects

BEST FOR

screen factor effects

AVOID WHEN

for observational data that were not generated by a valid designed experiment

› **Advanced contract · configuration, data, accessibility, and authority**

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DOE Main Effects



RF bias

Pressure

41.4



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ANALYTIC · DOE

DOE Interactions

DOE interaction plot

BEST FOR
factor interaction assessment

AVOID WHEN
for observational data that were not generated by a valid designed experiment

› **Advanced contract · configuration, data, accessibility, and authority**

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

DOE Interactions

+

−

● Pressure low ● Pressure high

41.3

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ANALYTIC · DOE

DOE Response Surface

DOE response surface

BEST FOR

quadratic two-factor response optimization

AVOID WHEN

for observational data that were not generated by a valid designed experiment

Advanced contract · configuration, data, accessibility, and authority

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Live example is ready.

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Canonical synthetic reference fixture; engineering conclusions require validated production data.

DOE Response Surface

+ - [] ↺ [] ⌵ ⌶

3.0

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RECIPE · SEMICONDUCTOR RECIPE

SPCMonitor

Monitor process stability, capability and violations with drill-down to manufacturing context.

BEST FOR

monitor a process parameter · detect process drift or control-rule violations

AVOID WHEN

when the process has no defensible baseline or chart family/subgroup definition

> Advanced contract · configuration, data, accessibility, and authority

Related capabilities

SPC I-MR

Capability Histogram

ECDF

Wafer Continuous

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SPCMonitor Canonical sample composition

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ANALYTICAL · SPC_I_MR

ANALYTICAL · CAPABILITY_HISTOGRAM

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RECIPE · SEMICONDUCTOR RECIPE

ExcursionDefenseLine

Triage an excursion with affected/control populations, spatial localization, commonality and genealogy evidence.

BEST FOR

build an excursion defense line · triage affected versus control material

AVOID WHEN

when affected/control population definitions are not yet governed or comparable

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

RCA Affected Control

Wafer Delta

RCA Commonality Ranking

RCA Commonality Matrix

RCA Genealogy Graph

RCA Evidence Matrix

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

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ExcursionDefenseLine

Canonical sample composition

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ANALYTICAL · RCA_AFFECTED_CONTROL

ANALYTICAL · WAFER_DELTA

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RECIPE · SEMICONDUCTOR RECIPE

FdcToolHealth

Diagnose equipment health from aligned sensor traces, envelopes, events and multivariate fingerprints.

BEST FOR

analyze FDC traces · check tool or chamber health

AVOID WHEN

when recipe-step alignment or sensor/event semantics are unavailable

➤ Advanced contract · configuration, data, accessibility, and authority

Related capabilities

- [FDC Recipe Step Trace](#)
- [FDC Golden Envelope](#)
- [FDC Multi Sensor](#)
- [FDC Chamber Fingerprint](#)
- [FDC PCA Scores](#)
- [FDC PCA Loadings](#)
- [FDC Equipment Event Overlay](#)

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FdcToolHealth Canonical sample composition

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ANALYTICAL · FDC_RECIPE_STEP_TRACE

ANALYTICAL · FDC_GOLDEN_ENVELOPE

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RECIPE · SEMICONDUCTOR RECIPE

LotWaferExplorer

Explore lot-to-wafer-to-die spatial behavior with synchronized maps, defects, profiles and records.

BEST FOR

explore wafers in a lot · inspect spatial metrology or defect signatures

AVOID WHEN

when wafer/die coordinates or wafer identity are unavailable

>

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

Wafer Continuous

Wafer Categorical

Lot Wafer Strip

Wafer Radial

Wafer Ring

Wafer Sector

Wafer Defect Clusters

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LotWaferExplorer

Canonical sample composition

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ANALYTICAL · WAFER_CONTINUOUS

ANALYTICAL · WAFER_CATEGORICAL

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RECIPE · SEMICONDUCTOR RECIPE

YieldLoss

Explain yield loss through Pareto/decomposition, bin/spatial signatures and commonality drill-down.

BEST FOR

analyze yield loss · rank fail bins or loss contributors

AVOID WHEN

when yield denominators, bin definitions or comparison populations are inconsistent

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

Yield Pareto

Bin Pareto

Yield Waterfall

Wafer Categorical

RCA Commonality Ranking

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YieldLoss

Canonical sample composition

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ANALYTICAL · YIELD_PARETO

ANALYTICAL · BIN_PARETO

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RECIPE · SEMICONDUCTOR RECIPE

PMEffectAnalysis

Compare pre/post-PM populations, chamber behavior, traces and process stability using one shared context.

BEST FOR

measure PM effect · compare chamber before and after maintenance

AVOID WHEN

when pre/post windows or maintenance-event timing cannot be established

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

RCA Affected Control

FDC Tool Chamber Compare

SPC I-MR

Box Distribution

FDC Equipment Event Overlay

FDC Chamber Fingerprint

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Live example is ready.

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PMEffectAnalysis

Canonical sample composition

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ANALYTICAL · RCA_AFFECTED_CONTROL

ANALYTICAL · FDC_TOOL_CHAMBER_COMPARE

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RECIPE · SEMICONDUCTOR RECIPE

ChamberMatching

Compare chamber signatures and distributions to detect mismatch and identify driving variables.

BEST FOR

match chambers · compare chambers across a toolset

AVOID WHEN

when chambers do not share comparable process/recipe populations

› Advanced contract · configuration, data, accessibility, and authority

Related capabilities

FDC Chamber Fingerprint

FDC Tool Chamber Compare

FDC PCA Scores

FDC PCA Loadings

Ridge Distribution

Wafer Comparison

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PreviewUsageDataInteractionsInspectCode

ChamberMatching Canonical sample composition

Confirm a Data mapping to restrict this composition to panels supported by your own development dataset.

ANALYTICAL · FDC_CHAMBER_FINGERPRINT

ANALYTICAL · FDC_TOOL_CHAMBER_COMPARE

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RECIPE · SEMICONDUCTOR RECIPE

RcaCockpit

Run structured root-cause analysis with populations, commonality, contribution, evidence and genealogy in one cockpit.

BEST FOR

perform RCA · investigate root cause or commonality

AVOID WHEN

when the workflow needs causal proof from observational association alone

Advanced contract · configuration, data, accessibility, and authority

Related capabilities

RCA Affected Control

RCA Commonality Ranking

RCA Commonality Matrix

RCA Contribution Waterfall

RCA Evidence Matrix

RCA Genealogy Graph

RCA Fault Tree

RCA Sankey

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RcaCockpit

Canonical sample composition

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Full Applications

Open a complete reference composition built from the department pattern and recipe authorities.

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SPC Control Center

A complete stability review composed from monitoring, dashboard, comparison and drill-down surfaces.

Is the critical-dimension process stable, and which observations need review?

critical dimension · control limits · review queue

Search records

Filter this application example

View

All records



critical dimension

control limits

review queue

Records

12

Synthetic reference fixture

Review focus

critical dimension

Domain context

Pattern

Monitoring

Primary analytical view

Chart and supporting records stay linked to the same filtered fixture.

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FDC Tool Health Center

A full operational surface for aligned traces, chamber context, event review and affected records.

Which tool or chamber signal changed around the observed equipment event?

pressure trace · tool/chamber · event review

Search records

Filter this application example

View

All records



pressure trace

tool/chamber

event review

Records

12

Synthetic reference fixture

Review focus

pressure trace

Domain context

Pattern

Monitoring

Primary analytical view

Chart and supporting records stay linked to the same filtered fixture.

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Excursion Investigation

An evidence-first investigation workspace for affected/control separation, commonality and supporting records.

What affected/control differences are worth investigating for this excursion?affected vs control · evidence ledger · hypothesis review

Search records

Filter this application example

View

All records

affected vs controlevidence ledgerhypothesis review

Primary analytical view

Chart and supporting records stay linked to the same filtered fixture.

Records

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Synthetic reference fixture

Review focus

affected vs control

Domain context

Pattern

Analysis Workspace

Open selected detail

EVIDENCE
CAVEATS